# Solutions for FIT3080 Applied Class on Intelligent Agents

## Exercise 1: Rationality

Describe the requirements of a rational vacuum agent for the case where each movement costs one point, assuming that once a square has been cleaned, it doesn't get dirty.

SOLUTION:
There are several sensible answers.

Such an agent requires a NoOp action to stop moving when it has cleaned all the dirty squares. Now, since the agent's percept doesn't say which squares have been previously cleaned, the agent must have some memory to say whether other squares have already been cleaned.

## Exercise 2: PEAS

For each of the following activities, give a PEAS description of the task environment, and characterize the task environment in terms of the environment types presented in the lecture slides.

(a) Playing soccer.

(b) Performing a high jump.

(c) Knitting a sweater.

SOLUTION:
There are several sensible answers.

(a) **Playing soccer.**

PEAS:

- Performance measure: scoring a goal, passing accurately.
- Environment: soccer field, spectators, operating conditions, referees, other players.
- Actuators: player's legs, head, hands (goal keeper).
- Sensors: player's eyes, ears, hands, legs, body, head.

Properties:

- Partially observable (the agent may not be able to see the whole field).
- Known.
- Multi agent.
- Stochastic (the ball may not go where it is kicked).
- Sequential (moves depend on previous moves).
- Dynamic (players move around while an agent is performing an action).
- Continuous (in terms of time, positions of players and of the ball).

(b) **Performing a high jump.**

PEAS:

- Performance measure: height of the jump.
- Environment: running-up track, jump area, high bar, spectators, judges.

- Actuators: jumper's legs.
- Sensors: jumper's eyes, ears, tactile.

Properties:

- Fully observable.
- Known.
- Single agent.
- Stochastic (effects of wind on the jump, grip of the surface).
- Sequential (the achieved height depends on the push-off).
- Static (the environment does not change).
- Continuous (in terms of time and positions of the agent).

(c) **Knitting a sweater.**

PEAS:

- Performance measure: quality of work, speed.
- Environment: needles, wool, knitting pattern.
- Actuators: knitter's hands.
- Sensors: knitter's eyes, hands.

Properties:

- Fully observable.
- Known.
- Single-agent.
- Stochastic (the agent may drop a stitch).
- Sequential (a stitch depends on the previous work).
- Static (the environment doesn't change).
- Discrete (each stitch is separate).

**Exercise 3:**

For each task in Exercise 2, consider the following types of agents, and for each type, describe what the agent can do, and how it differs from simpler agents: Simple reflex, Model based, Goal based and Utility based.

In addition, for each task, define *one* performance measure for which all the agents would be rational. Justify your answers.

SOLUTION:
Other solutions are possible.

(a) **Playing soccer.** Let us say that the percepts and actions are the same for all types of agents:

- Percepts: whether agent is in possession of the ball, coordinates in the field, distance from and direction of the goal, and distance from each other player.
- Actions: kick the ball (+ direction), run without the ball (+ direction), and run with the ball (+ direction). All the agents can pass the ball to another agent who is closer to the goal or kick towards the goal.

All the agents would be rational according to the performance measure **Is the ball closer to the goal?**, as they all have the capacity to perceive the goal, and kick the ball in that direction.

- **Simple reflex** – this agent has rules that tell it how to kick, e.g., "If you get the ball and you are the closest to the goal, turn towards the goal and kick". Note that this agent needs to update all its percepts before it takes action.
- **Model based** – this agent has a model of how the world evolves as a result of its actions and the actions of other agents, which enables the agent to deal with parts of the world it can't see. The agent kicks towards the goal or another player based on its model of the world, but it can't update all its percepts, as the environment is only partially observable.
- **Goal based** – this agent has an idea of how the world will change as a result of its actions, and can adjust its actions depending on what it wants to achieve (just knowing what the world looks like is not enough to adjust your actions). This agent can make a plan where it passes the ball to another agent or kicks towards the goal – the plan may include what other agents will do.
- **Utility based** – this agent incorporates information about the quality of the result of its actions, e.g., how close he can get the ball to the goal.

(b) **Performing a high jump.**

- Percepts: height of the bar, distance to the bar, ground conditions (e.g., slippery).
- Actions: run and jump.

All agents are rational under this performance measure: **whether the bar was cleared**, as they all have the capacity to perceive the bar, and attempt to clear it.

- **Simple reflex** – the rule may be "If distance from bar $> X$ run, else jump as hard as you can".
- **Model based** – if the above are the only relevant conditions, there is no need for a model, because the environment is fully observable. But the model can still be used to plan actions.
- **Goal based** – like the model-based agent.
- **Utility based** – in this case, the agent can maximize utility by adjusting the expended effort.

(c) **Knitting a sweater.**

- Percepts: position of knitting needles (row number and stitch number), remaining yarn, instructions for a complete sweater, instruction number last executed.
- Actions: knit, purl, reduce, increment, cast on (the knitting needle), cast off, sew together the parts.

All agents are rational under this performance measure **Is the sweater finished?**, as they can all knit, and they can perceive whether the sweater is complete. However, a simple reflex agent would require rules for every step of knitting in order to chose the next action.

- **Simple reflex** – the rules must be very detailed (stitch by stitch for the entire sweater).

- **Model based** – this agent has a model of what is the outcome of a knitting action (e.g., you have a bigger section of knit material), but since the environment is fully observable, the model is not really required.
- **Goal based** – this agent would be able to make a plan to knit the different portions of a sweater using operators that combine several knitting actions.
- **Utility based** – this agent would be able to judge the quality of the sweater and the knitting speed.

**Exercise 4: Vacuum-cleaner World and Agent (Python code)**

(a) Install a Python environment on your device. You can use Anaconda, Jupyter Notebook, or any other Python IDE of your choice.

(b) Define Percepts and Actions using a suitable data structure in Python:

- Percepts: Represent the state of the vacuum cleaner'sworld, e.g.,["A", "Dirty"] or ["B", "Clean"].
- Actions: Define possible actions the vacuum cleaner can take, e.g.,"Left", "Right", "Vacuum".

(c) Write a program that implements a rational agent in the Vacuum-cleaner World. Your program should:

- Read the current percept (location and state) of the vacuum cleaner.
- Determine the appropriate action to take based on the current percept.
- Execute the action and update the percept accordingly.

SOLUTION:
See associated Python program.